KRITIKA GUPTA

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SUMMARY

MS Computer Science graduate (GPA 3.77/4.0) with hands-on Python backend, REST API, database, and applied ML experience. Built full-stack and LLM-powered web apps; co-authored a peer-reviewed IEEE paper on deep learning. Seeking an entry-level software or ML engineering role.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, JavaScript
Backend & Web: FastAPI, SQLAlchemy, Flask, Django, REST APIs, React, Vite, Axios, Leaflet.js, HTML/CSS
AI & ML: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Deep Learning (CNN, GRU, Attention), LLM APIs (Google Gemini, DeepSeek via Ollama), Tool/Function Calling, Prompt Engineering
Data, Cloud & Tools: PostgreSQL, MySQL, Oracle, DynamoDB, AWS (DynamoDB, boto3, IAM), Docker, Git, Azure and Kubernetes (coursework), Tableau, Agile/Scrum, Code Review

EXPERIENCE

Chicago Education Advocacy Cooperative (ChiEAC) - Data Science Fellow / AI Data Analyst (Remote) **Jul 2026 - Present**
- Applying data analysis and machine learning methods to support education-equity programs.

ThinkNext Technologies - Software Engineering Intern **Jun 2022 - Jul 2022**
- Built and maintained Python/Django backend services for a production inventory management system.
- Optimized database interactions across MySQL and Oracle; collaborated with the front-end team and took part in code review.

GTB Institute - Python Programming Intern **Jul 2021 - Sep 2021**
- Used Pandas and NumPy for data analysis and visualization; led a team of 3 to deliver an airline management system and received formal recognition for leadership.

CMU Dining Services - Team Member **Oct 2025 - May 2026**
- Delivered efficient customer service and collaborated with a diverse team in a fast-paced environment.

PROJECTS

Regime-Aware Deep Learning for Predictive Maintenance (Hybrid SAMB-GRU) **2026**
Published in IEEE Xplore (MLISE 2026)
- Co-developed a Conv1D → BiGRU → attention model for Remaining Useful Life prediction on NASA C-MAPSS, with K-Means regime clustering (k = 6) and per-regime normalization to isolate degradation signal.
- Used engine-level train/validation splits to prevent leakage; reached R^2 of 0.87–0.88 and RMSE of 13.5–15.4 across FD001–FD004.

CommunityFlow Chicago - AI Community Intelligence Platform **2026**
FastAPI · SQLAlchemy · PostgreSQL · React/Vite · Leaflet · Gemini API
- Built a full-stack dashboard for evaluating Chicago neighborhoods, with an interactive Leaflet map, filters (transit score, parks, libraries), and CSV export.
- Integrated Gemini for a conversational advisor and AI-generated reports; added client-side caching and backend retry logic with database fallbacks to stay within free-tier rate limits.

Agentic AI Smart City Assistant (Capstone) **2026**
Flask · DeepSeek-V3.1 via Ollama · Open-Meteo
- Built a Flask app where an LLM agent interprets natural-language weather queries, calls an Open-Meteo-backed tool, and returns a synthesized answer, with intent detection and documented REST endpoints.

URL Shortener with Cloud Integration **2024**
FastAPI · AWS DynamoDB · JavaScript
- Built a FastAPI + DynamoDB URL shortener with custom codes, TTL expiry, click analytics, tags, and QR-code generation, plus a vanilla JavaScript frontend.

PUBLICATION

Gupta, K., Barot, N., & Gargees, R. S. (2026). *Regime-Aware Deep Learning Framework for Predictive Maintenance in Industrial Systems.* MLISE 2026. DOI: 10.1109/MLISE70044.2026.11607667

EDUCATION

Central Michigan University - MS, Computer Science (GPA 3.77/4.0) **May 2026**
DAV University, India - BTech, Computer Science (GPA 3.07/4.0) **Jun 2023**

CERTIFICATIONS & HONORS

Intro to Generative AI (Google Cloud) · Azure Fundamentals · Foundations of Agents (Hugging Face) · Deep Learning Fundamentals (NVIDIA)
Departmental Scholarship (CMU) · Leadership Award (GTB Institute) · Bronze Medal, National Science Olympiad